

The Emergent State Machine: A Formal Framework for Interpretable Situational Reasoning

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Abstract

The Emergent State Machine (ESM) defines a deterministic, turn-based framework for interpretable reasoning over evolving situations. While prior work establishes the architectural structure of the ESM, the present paper formalizes the mathematical properties that characterize its operation.

We define the signal, state, and projection spaces underlying the system and specify the mappings that govern each turn. From this structure, we establish a set of invariants, including reconstructive state evolution, fixed state space, signal-mediated persistence, and deterministic decision boundaries. These properties imply that system trajectories are uniquely determined, exactly reproducible, and geometrically consistent across time.

Unlike systems that propagate hidden state or rely on evolving representations, the ESM constructs state anew at each turn from explicitly defined signals, with persistence encoded through signal construction rather than state carryover. Decision-making is governed by a deterministic gating mechanism that controls both the invocation and outcome of policy evaluation.

We show that these invariants induce a class of systems whose behavior admits interpretation through dynamical systems and control theory, while remaining fully inspectable and modular. The contribution of this work is a formal characterization of the ESM as a constrained class of systems defined by structural invariants rather than implementation details.

1 Contributions

This paper makes the following contributions:

- **Formalization of the ESM structure.** We define the signal space, state space, and projection space, along with the deterministic mappings that govern turn-based evolution.

- **Reconstructive state evolution.** We establish that state is constructed at each turn from signals rather than propagated from prior state, with persistence implemented explicitly through signal construction.
- **Invariant state space and geometric consistency.** We show that the state space is fixed across time, ensuring that trajectories admit consistent geometric interpretation.
- **Deterministic decision boundaries.** We formalize relevance gating as a deterministic mechanism that governs both the invocation and outcome of policy evaluation.
- **Signal-mediated composition.** We prove that interaction between ESMs occurs exclusively through signals, preserving independence of state construction and enabling modular composition.
- **Replayability and auditability.** We demonstrate that system trajectories are uniquely determined and exactly reproducible given identical inputs and definitions.
- **Interpretation via dynamical systems and control theory.** We show that the invariants of the ESM induce behavior that can be analyzed using geometric and control-theoretic frameworks, while remaining defined by structural constraints rather than those frameworks.

2 Introduction

The Emergent State Machine (ESM) defines a deterministic, turn-based framework for reasoning over evolving situations. A complete architectural description is provided in prior work [4], which introduces the system components, execution flow, and practical instantiations. The formal specification is available in [5].

The present work takes a complementary perspective. Rather than focusing on the architectural description and instantiation of the system, we formalize the mathematical structure implied by the architecture and establish the invariants that govern its operation. The focus is therefore on the properties that define the class of systems induced by the ESM.

At a high level, the ESM organizes reasoning into discrete turns. Each turn transforms observations into signals, constructs an interpretable state, projects that state into policy-relevant coordinates, evaluates relevance through explicit gating, and conditionally applies policy. This paper treats that process as a composition of deterministic mappings between formally defined spaces.

From this formulation, several structural properties emerge. State is not propagated across time but reconstructed at each turn from signals. Persistence is implemented explicitly through signal construction rather than implicit state carryover. The state space is fixed, ensuring that trajectories evolve within a stable geometric representation. Decision-making is governed

by deterministic gating, which controls both whether policy is invoked and what outcome is produced.

These properties define a constrained class of systems with distinct behavior. Trajectories are uniquely determined by inputs, exactly reproducible, and geometrically consistent across time. Interaction between systems occurs exclusively through signals, preserving independence of state construction and enabling modular composition.

While the resulting behavior admits interpretation through dynamical systems and control theory, these perspectives are not foundational. The defining characteristics of the ESM are the invariants that govern its construction and operation; geometric and control-theoretic views arise as consequences of these constraints rather than as assumptions.

This paper assumes familiarity with the architectural description but remains self-contained at the level of formal definitions and properties. Readers seeking implementation detail and system intuition are referred to the architecture specification; readers seeking formal guarantees may treat the present work as primary.

The ESM is defined not by its components, but by the invariants it enforces.

3 Formal Structure

The Emergent State Machine (ESM) operates as a discrete-time system whose evolution proceeds through successive turns. Each turn T_t produces a fully constructed and committed state x_t , which serves as the authoritative representation of the system at time t .

$$\sigma : \mathcal{O} \rightarrow \mathcal{S}, \quad C : \mathcal{S} \rightarrow \mathcal{X}, \quad P : \mathcal{X} \rightarrow \mathcal{Y}, \quad G : \mathcal{Y} \rightarrow \{0, 1\}, \quad \pi : \mathcal{Y} \rightarrow \mathcal{R}. \quad (1)$$

3.1 Turn Structure

Each turn T_t is defined as:

$$T_t = (o_t, S_t, x_t, y_t, g_t, r_t) \quad (2)$$

where:

- o_t denotes the observations at time t ,
- $S_t = \sigma(o_t)$ denotes the derived signals,
- $x_t = C(S_t)$ denotes the constructed state,
- $y_t = P(x_t)$ denotes the projection into decision coordinates,
- $g_t = G(y_t)$ denotes the relevance gating outcome,

- r_t denotes the policy outcome, defined conditionally on g_t (see §3.3).

The state x_t is not a primitive input but a deterministic construction from signals via C , and is recorded as part of the turn once constructed.

The causal structure of the turn is given by the deterministic transformation sequence:

$$o_t \rightarrow S_t \rightarrow x_t \rightarrow y_t \rightarrow g_t \rightarrow r_t \quad (3)$$

ensuring that all outcomes arise from explicitly defined transformations.

Signal Construction Signals are constructed deterministically from the admitted observation set:

$$S_t = \sigma(o_t), \quad (4)$$

where σ denotes the explicitly defined signal construction mapping. σ is fixed and versioned as part of the system definition. Each signal represents a derived or directly encoded quantity of interest, including transformations such as thresholds, aggregations, ratios, or categorical encodings.

Signal construction is fully specified and inspectable; no information enters the system except through o_t , and all elements of S_t are reproducible from the observation set under σ .

3.2 State Construction and Projection

State is constructed deterministically from signals:

$$x_t = C(S_t), \quad (5)$$

where C denotes the explicit, versioned construction rule.

Projection maps state into decision coordinates:

$$y_t = P(x_t), \quad (6)$$

without introducing new semantic content.

The projection space $\mathcal{Y}$ defines the domain on which relevance gating operates, with G establishing the coordinates used for decision activation.

3.3 Relevance Determination and Policy

Relevance gating determines whether policy evaluation is invoked for the projected state. Formally,

$$g_t = G(y_t). \quad (7)$$

If $g_t = 1$, the policy π is applied to y_t ; if $g_t = 0$, no policy output is produced. Thus, gating defines the decision-relevant subset of projection space on which policy operates.

3.4 State Evolution

System evolution proceeds through successive turns as repeated state construction from the observation set at each turn. The committed state at time t is given by

$$x_t = C(S_t). \quad (8)$$

Trajectories arise as the resulting sequences

$$\{x_1, x_2, \dots\}, \quad \{y_1, y_2, \dots\}, \quad \{T_1, T_2, \dots\}. \quad (9)$$

Evolution occurs only at discrete turn boundaries. Any influence of prior turns must enter through the admitted observations and derived signals available at the current turn; it is not introduced through direct recursive state propagation.

4 Properties

The formal structure of the Emergent State Machine (ESM) induces several invariants that hold across all implementations of the framework.

4.1 Deterministic Execution

Given fixed definitions of signal extraction, state construction, projection, gating, and policy, the system is deterministic with respect to the admitted observation set o_t :

$$o_t \Rightarrow r_t \quad (10)$$

is uniquely determined.

Equivalently, the composed transformation

$$o_t \mapsto S_t \mapsto x_t \mapsto y_t \mapsto g_t \mapsto r_t \quad (11)$$

is deterministic for all t , as each mapping is explicitly defined and versioned.

4.2 Turn-Level Completeness

Each turn $T_t = (o_t, S_t, x_t, y_t, g_t, r_t)$ constitutes a complete and self-contained reasoning unit. All quantities required to reconstruct the outcome r_t are contained within the turn and its associated definitions.

This implies that every turn is independently auditable and replayable.

4.3 Separation of Representation and Decision

The projection P separates situational representation from decision evaluation:

$$y_t = P(x_t) \tag{12}$$

All decision logic (gating and policy) operates exclusively on y_t , not directly on x_t or S_t . This enforces a strict boundary between:

- representation (x_t)
- decision coordinates (y_t)

4.4 Conditional Invocation of Policy

Policy is not universally applied at every turn. Instead, invocation is governed by gating:

$$g_t = G(y_t). \tag{13}$$

Thus, decision-making is conditionally activated, yielding:

$$r_t = \begin{cases} \pi(y_t) & \text{if } g_t = 1 \\ \emptyset & \text{if } g_t = 0 \end{cases} \tag{14}$$

This establishes a two-stage decision structure: relevance determination followed by action selection.

4.5 Locality of Evolution

State evolution is locally governed at each turn through reconstructive state construction:

$$x_t = C(S_t). \tag{15}$$

No global optimization or trajectory-level objective is required for state evolution.

Trajectories emerge as a consequence of sequential local construction:

$$\{x_1, x_2, \dots\}.$$

This implies that system behavior is constrained by the observations and signals available at each turn. Any apparent path dependence arises only through information carried forward

via admitted observations or derived signals, not through direct recursive dependence on prior state.

4.6 Explicit Causal Trace

Each outcome r_t admits a complete causal trace:

$$o_t \rightarrow S_t \rightarrow x_t \rightarrow y_t \rightarrow g_t \rightarrow r_t. \quad (16)$$

This trace is fully inspectable and sufficient to identify the origin of any outcome within the transformation chain.

4.7 Composability via Handoff

Layered ESMs compose through explicit state handoff:

$$x_1^{(d)} = H\left(x_{\text{handoff}}^{(u)}, o_1^{(d)}\right), \quad (17)$$

where H is a deterministic transformation.

This ensures that composition preserves determinism and auditability, with no hidden coupling between systems; handoff artifacts do not constitute state but serve as inputs to signal construction in the downstream system.

5 Proof Sketches

The following proof sketches show that the properties in Section 4 follow directly from the formal structure of the ESM.

5.1 Deterministic Execution

Proof Sketch. All component mappings are assumed to be total over their declared domains. Therefore, for any given input o_t , the sequence of transformations leading to r_t is fully determined.

The turn structure is a composition of explicitly defined deterministic transformations:

$$o_t \mapsto S_t \mapsto x_t \mapsto y_t \mapsto g_t \mapsto r_t. \quad (18)$$

More explicitly,

$$S_t = \sigma(o_t), \quad (19)$$

$$x_t = C(S_t), \quad (20)$$

$$y_t = P(x_t), \quad (21)$$

$$g_t = G(y_t), \quad (22)$$

and

$$r_t = \begin{cases} \pi(y_t) & \text{if } g_t = 1 \\ \emptyset & \text{if } g_t = 0 \end{cases} \quad (23)$$

Since each mapping is deterministic and total, each stage produces a unique output for a given input. Therefore, the composition of these mappings also produces a unique output r_t for each o_t .

Since each mapping is also explicitly defined and versioned, the composed mapping from o_t to r_t is deterministic. Therefore, identical observations under identical definitions yield identical outcomes. $\square$

5.2 Turn-Level Completeness

Proof Sketch. A turn is defined as

$$T_t = (o_t, S_t, x_t, y_t, g_t, r_t). \quad (24)$$

This tuple contains the observations, derived representational forms, gating result, and final outcome associated with the turn. Together with the explicit definitions of the mappings σ , C , P , G , and π , the tuple is sufficient to reconstruct the full causal chain leading to r_t .

Hence, each turn is complete as an auditable reasoning unit. $\square$

5.3 Separation of Representation and Decision

Proof Sketch. By construction,

$$y_t = P(x_t), \quad (25)$$

and both gating and policy operate only on projected state:

$$g_t = G(y_t), \quad r_t = \begin{cases} \pi(y_t) & \text{if } g_t = 1 \\ \emptyset & \text{if } g_t = 0 \end{cases} \quad (26)$$

Neither x_t nor S_t enters the decision boundary directly. Therefore, the framework enforces a structural separation between situational representation and decision evaluation. $\square$

5.4 Conditional Invocation of Policy

Proof Sketch. Policy is evaluated only through the branch condition imposed by gating:

$$g_t = G(y_t). \quad (27)$$

If $g_t = 0$, the outcome is explicitly defined as $\emptyset$. If $g_t = 1$, policy is invoked:

$$r_t = \begin{cases} \pi(y_t) & \text{if } g_t = 1 \\ \emptyset & \text{if } g_t = 0 \end{cases} \quad (28)$$

Thus, the architecture distinguishes deterministically between relevance determination and policy execution. $\square$

5.5 Locality of Evolution

State evolution is locally governed at each turn through reconstructive state construction:

$$x_t = C(S_t), \quad (29)$$

without reliance on global optimization or trajectory-level objectives.

Trajectories emerge as a consequence of sequential local construction:

$$\{x_1, x_2, \dots\}.$$

This implies that system behavior is constrained by the observations and signals available at each turn. Any apparent path dependence arises only through information carried forward via admitted observations or derived signals, not through direct recursive dependence on prior state.

Proof Sketch. This follows directly from the construction rule $x_t = C(S_t)$ and the definition of $S_t = \sigma(o_t)$, which depend only on observations at time t . No dependence on prior states is introduced. $\square$

5.6 Explicit Causal Trace

Proof Sketch. The architecture defines the causal ordering

$$o_t \rightarrow S_t \rightarrow x_t \rightarrow y_t \rightarrow g_t \rightarrow r_t. \quad (30)$$

Each term in this chain is explicitly represented in the turn record and linked by deterministic mappings. Therefore, for any realized outcome r_t , the full sequence of intermediate representational and decision stages is recoverable.

This yields an explicit causal trace for every turn. $\square$

5.7 Composability via Handoff

Proof Sketch. For layered ESMs, downstream initialization is defined by

$$x_1^{(d)} = H\left(x_{\text{handoff}}^{(u)}, o_1^{(d)}\right), \quad (31)$$

where H is deterministic and explicit.

All state construction, at initialization and during operation, occurs exclusively through signals; handoff artifacts serve only as inputs to signal construction and do not constitute state.

Thus, inter-ESM coupling occurs only via defined handoff artifacts at turn boundaries. Because H is explicit and deterministic, composition preserves auditability and determinism without introducing hidden state coupling. $\square$

6 Implications for Control and Geometry

The formal structure of the Emergent State Machine (ESM) admits a natural interpretation within control theory and geometric frameworks. These interpretations clarify how the system relates to classical formulations while highlighting key structural distinctions.

6.1 Control-Theoretic Interpretation

The ESM can be understood as a discrete-time controlled system evolving through committed states constructed at each turn:

$$x_t = C(S_t). \quad (32)$$

This formulation makes explicit that:

- Control is *event-conditional*, governed by $G(y_t)$
- Decision structure is defined in projection space $\mathcal{Y}$, not directly in $\mathcal{X}$
- System evolution proceeds through deterministic state construction at each turn
- Inter-system influence occurs only through explicit handoff H

6.2 Geometric Interpretation

State evolves in a representation space $\mathcal{X}$, with projection defining a mapping

$$y_t = P(x_t), \quad (33)$$

where $\mathcal{Y}$ is the policy-relevant coordinate space.

This induces a separation between $\mathcal{X}$, the situational representation space, and $\mathcal{Y}$, the decision-coordinate space.

Gating defines a subset of decision space:

$$\mathcal{Y}_G = \{y \in \mathcal{Y} \mid G(y) = 1\}, \quad (34)$$

within which policy is evaluated.

System behavior can be understood as trajectories

$$\{x_1, x_2, \dots\} \subset \mathcal{X}, \quad \{y_1, y_2, \dots\} \subset \mathcal{Y}, \quad (35)$$

with policy applied only when trajectories lie within the gated region $\mathcal{Y}_G$.

Relevance gating determines whether policy evaluation is invoked; it does not control state construction or trajectory evolution. The system constructs and monitors x_t and y_t at every turn regardless of whether $g_t = 1$.

Thus, a trajectory may evolve without triggering policy evaluation, reflecting conditions that require monitoring rather than intervention.

Projection P acts as a coordinate transformation that preserves decision-relevant structure while discarding irrelevant detail. Changes to P redefine the geometry of $\mathcal{Y}$ without altering the underlying representation in $\mathcal{X}$.

6.3 Interpretive Consequences

The properties above jointly imply that control in ESM is not continuous but structurally mediated: evaluation occurs only when relevance conditions are satisfied, and decisions operate on projected representations rather than raw state. This separation between representation, projection, and conditional action defines a form of governed, event-triggered control that differs from continuous feedback formulations.

These consequences follow directly from the formal definitions and characterize how system behavior arises within the framework, including its geometric structure, interpretive formulation, decision structure, and inter-system composition.

7 Related Work

We now shift from internal characterization to external positioning. The following section situates these properties relative to established domains, relating the components of the framework to existing areas of study without introducing new mechanisms.

7.1 Control and Decision Frameworks

Classical control theory and decision processes study how systems select actions based on state in order to achieve desired outcomes under constraints [2]. In standard formulations, a control law or policy maps state directly to action:

$$u_t = \kappa(x_t).$$

Frameworks such as Markov Decision Processes (MDPs) formalize this interaction by defining policies over state spaces, often with an emphasis on optimality, reward maximization, or stability under known dynamics.

In these approaches, the state representation is typically assumed to be given or inferred, and decision-making is applied uniformly at each step. Analysis focuses on how policies interact with system dynamics to produce desired behavior over time.

The Emergent State Machine (ESM) differs by making the construction of state explicit and by separating the decision process into distinct stages. State is not assumed; it is deterministically constructed from decision-relevant signals at each turn. This constructed state is then projected into a decision space, where relevance is evaluated before any policy is applied:

$$g_t = G(y_t), \quad r_t = \begin{cases} \pi(y_t) & \text{if } g_t = 1 \\ \emptyset & \text{if } g_t = 0 \end{cases}$$

This introduces an explicit distinction between determining when a decision should be made and what decision should be taken. Policy is not continuously or implicitly applied, but conditionally invoked based on inspectable criteria defined over the projected state. As a result, the decision process itself becomes structured and analyzable, rather than uniformly embedded within system evolution.

In practice, this corresponds to systems where decisions are triggered only when explicitly defined conditions are met, and where both the inputs to the decision and the conditions for invoking it are constructed and inspectable rather than assumed or inferred.

Classical control and decision frameworks provide powerful tools for analyzing and optimizing system behavior. ESM complements these approaches by retaining the policy-over-state formulation while extending it with explicit state construction and relevance gating, enabling direct inspection of both decision inputs and decision activation within a unified, deterministic structure.

7.2 Dynamical Systems and Cybernetics

Classical cybernetics and dynamical systems theory study how systems evolve over time under feedback, with particular emphasis on stability, equilibrium, and attractor behavior [1, 7].

System behavior is typically characterized through state trajectories governed by explicit dynamics, often formulated in continuous time or as discretizations of underlying differential equations. Analysis focuses on how feedback structures regulate evolution, giving rise to stable regimes, oscillations, or more complex dynamical patterns.

These frameworks generally assume that the system state is either directly measurable or implicitly defined within a state-space model. In many cases, the state representation is latent or only partially observable, and reasoning about system behavior requires analysis of the governing equations rather than direct inspection of state at each step.

The Emergent State Machine (ESM) adopts a discrete-time formulation in which system evolution proceeds through a sequence of explicit, committed states $x_1, x_2, \dots$, each constructed at a turn boundary from observed and derived signals. Rather than treating state as inferred or hidden, ESM defines state as an inspectable, decision-relevant representation of the current situation. Each transition is produced by a deterministic process, yielding a trajectory that can be replayed and audited.

The ESM framework shares the emphasis on trajectory-based analysis and feedback-governed evolution, but differs in how state and dynamics are constructed. Evolution proceeds in discrete, committed steps, where each transition results from a fully constructed state, an explicit projection, and a deterministic policy evaluation.

In this formulation, stability is defined relative to policy-governed desirable regions in state space, rather than solely as mathematical equilibrium. A system is stable if its trajectory remains within, or returns to, these regions under the applied policy and gating conditions. In practice, this corresponds to systems evaluated by whether they stay within acceptable operating bounds across successive decisions, rather than whether they converge to a fixed point.

Feedback in ESM is explicit and structured: signals derived from observations influence state construction, projection determines decision relevance, and policy governs action. This contrasts with systems in which feedback is embedded implicitly within learned parameters or continuous dynamics. Making feedback pathways explicit enables direct inspection of how specific signals influence system evolution.

Classical dynamical systems and cybernetics provide tools for analyzing system evolution, feedback, and stability over time. ESM complements these approaches by providing a structure in which system evolution is represented as discrete, inspectable trajectories over explicitly constructed state, preserving dynamical behavior while enabling direct auditability of each transition.

7.3 Interpretability and Human-in-the-Loop Systems

Interpretability and human-in-the-loop systems focus on enabling humans to understand, trust, and effectively interact with automated decision systems. Prior work in explainable AI (XAI)

often centers on explaining the outputs of complex or opaque models, using techniques such as feature attribution, surrogate models, or post hoc explanations. In parallel, research on situation awareness emphasizes the importance of presenting system state in a way that supports human perception, comprehension, and projection in dynamic environments [3].

These approaches typically operate downstream of model execution, seeking to make internal reasoning legible after decisions have been produced. As a result, interpretability is often treated as an auxiliary layer applied to systems whose internal representations and decision processes remain partially or fully opaque.

The Emergent State Machine (ESM) takes a different approach by making interpretability a property of the system’s construction rather than an added capability. State is explicitly defined as a collection of decision-relevant variables derived from observable signals, and each step in the decision process—state construction, projection, gating, and policy evaluation—is inspectable. This shifts interpretability from explaining outcomes to exposing the structure of decision-making itself.

This formulation aligns with the goals of situation awareness while altering where those properties arise. Rather than reconstructing system state for human interpretation, ESM presents state as the primary object of reasoning, defined in terms of domain-relevant variables that directly support perception, comprehension, and projection over time.

Human integration in ESM is supported through this explicit structure. Because state and decision pathways are defined in terms of meaningful, domain-relevant variables, human operators can directly evaluate, modify, or extend system behavior without requiring access to latent representations or model internals.

In practice, this corresponds to systems where a user can trace a decision from observed inputs through state and policy, and intervene by adjusting definitions or thresholds rather than interpreting model outputs.

While existing interpretability techniques remain valuable in contexts involving complex learned models, ESM complements these approaches by enabling systems in which interpretability is inherent to the decision process, supporting direct human oversight and governance.

7.4 Uncertainty and Decision-Making

Decision theory and probabilistic frameworks address how systems make decisions under uncertainty, often modeling incomplete information, stochastic dynamics, and trade-offs between competing outcomes. Approaches such as Bayesian inference, expected utility theory, and Partially Observable Markov Decision Processes (POMDPs) incorporate uncertainty directly into the decision process, typically by assigning probabilities to states, observations, or outcomes [2].

In these formulations, uncertainty is often embedded within the state representation or policy itself, and decision-making involves optimizing expected outcomes given probabilistic

beliefs about the system. This enables principled handling of risks and incomplete information, but can make the internal basis for decisions difficult to inspect directly, particularly when probabilities are learned or estimated from data.

The Emergent State Machine (ESM) takes a different approach by representing uncertainty explicitly as part of the constructed state, rather than requiring it to be embedded within a probabilistic model or policy. Confidence, ambiguity, or data quality can be included as decision-relevant variables alongside other signals, allowing uncertainty to be surfaced and acted upon directly within the decision process.

This separates the representation of uncertainty from the mechanism of decision-making. Policy operates over explicit representations of uncertainty rather than over inferred probability distributions, and does not require optimization over expected values unless such behavior is explicitly defined. As a result, uncertainty becomes inspectable and structurally integrated into decision conditions, rather than implicitly encoded within model parameters.

In practice, this corresponds to systems where uncertainty is exposed as a first-class component of state and can directly influence whether and how decisions are made, including whether a decision should be deferred, escalated, or modified based on confidence or data quality.

Classical decision theory and probabilistic methods provide rigorous tools for reasoning under uncertainty and for optimizing decisions in stochastic environments. ESM complements these approaches by preserving the ability to reason about uncertainty while expressing it through explicit, inspectable state variables that directly inform gating and policy evaluation within a deterministic execution model.

7.5 Governance, Auditability and System Oversight

Governance and auditability in decision systems focus on ensuring that system behavior can be monitored, understood, and evaluated over time. Frameworks such as the NIST AI Risk Management Framework emphasize transparency, traceability, and accountability [6], particularly in high-stakes or regulated environments. These approaches require the ability to reconstruct how decisions were made, assess compliance with policies, and evaluate system behavior under changing conditions.

In many systems, governance is implemented through logging, monitoring, or post hoc analysis of decisions. However, when internal representations or decision processes are opaque, reconstructing the reasoning behind specific outcomes can be difficult or incomplete.

The Emergent State Machine (ESM) incorporates governance directly into the structure of the system through its turn-based formulation and explicit state construction. Each decision occurs within a defined turn that records observations, signals, constructed state, projection, gating outcome, and policy evaluation, producing a complete and inspectable record of system behavior at each step.

Because state and decision pathways are explicitly defined, changes to system behavior, such as adjustments to signal definitions, gating conditions, or policy rules, can be versioned and analyzed over time. This enables direct comparison of system behavior across configurations and supports reproducibility of past decisions without requiring reconstruction of hidden processes.

In practice, this corresponds to systems where every decision can be traced from inputs through state and policy, and where changes to system logic can be audited and replayed as part of the system’s normal operation.

Governance frameworks provide essential guidance for managing risk, ensuring accountability, and maintaining trust in automated systems. ESM complements these approaches by embedding auditability and traceability directly into the execution model, enabling governance as an inherent property of system operation rather than an external layer.

Summary

The preceding comparisons situate the framework relative to established domains without introducing new mechanisms. Each component, state construction, projection, gating, policy, and evolution, corresponds to recognizable structures in prior work, while remaining explicitly defined and inspectable within a unified system.

Unlike approaches that distribute semantics, decision logic, and system evolution across loosely coupled processes, the framework consolidates these elements within a single deterministic structure. This enables direct analysis of behavior through state trajectories, explicit decision coordinates, and reproducible turn-level transitions.

The result is a system in which interpretation, decision-making, and evolution are operationally integrated within a single deterministic structure. This integration is formally consistent with existing theoretical perspectives while preserving transparency at the level of execution.

8 Conclusion

We introduced the Emergent State Machine (ESM) as a deterministic framework for interpretable situational reasoning and governed decision-making. The architecture is defined by a discrete-time execution model in which each turn constructs a coherent state from observations and signals, projects that state into decision-relevant coordinates, evaluates relevance through explicit gating, and applies policy when activated.

This formulation separates interpretation from decision authority while maintaining a continuous, inspectable chain from input to outcome. State is not latent or inferred implicitly, but explicitly constructed and versioned. Projection defines the space in which decisions are made. Gating determines when decisions are invoked. Policy operates only on projected state, ensuring that all actions are grounded in defined coordinates.

System behavior emerges as a sequence of committed states over time. These trajectories are analyzable, reproducible, and directly attributable to formal components of the framework. Inter-system composition occurs through explicit handoff functions, preserving modularity without introducing hidden dependencies.

The framework supports transparent integration of model-assisted interpretation while maintaining deterministic control at the decision boundary. By structuring state change explicitly at the level of the turn and preserving full traceability of state transitions, the system enables auditability, replay, and governance without requiring post hoc reconstruction.

ESM provides a general foundation for systems in which decisions must be interpretable, reproducible, and governed. Its explicit treatment of state, projection, and policy offers a basis for both practical implementation and formal analysis across domains.

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